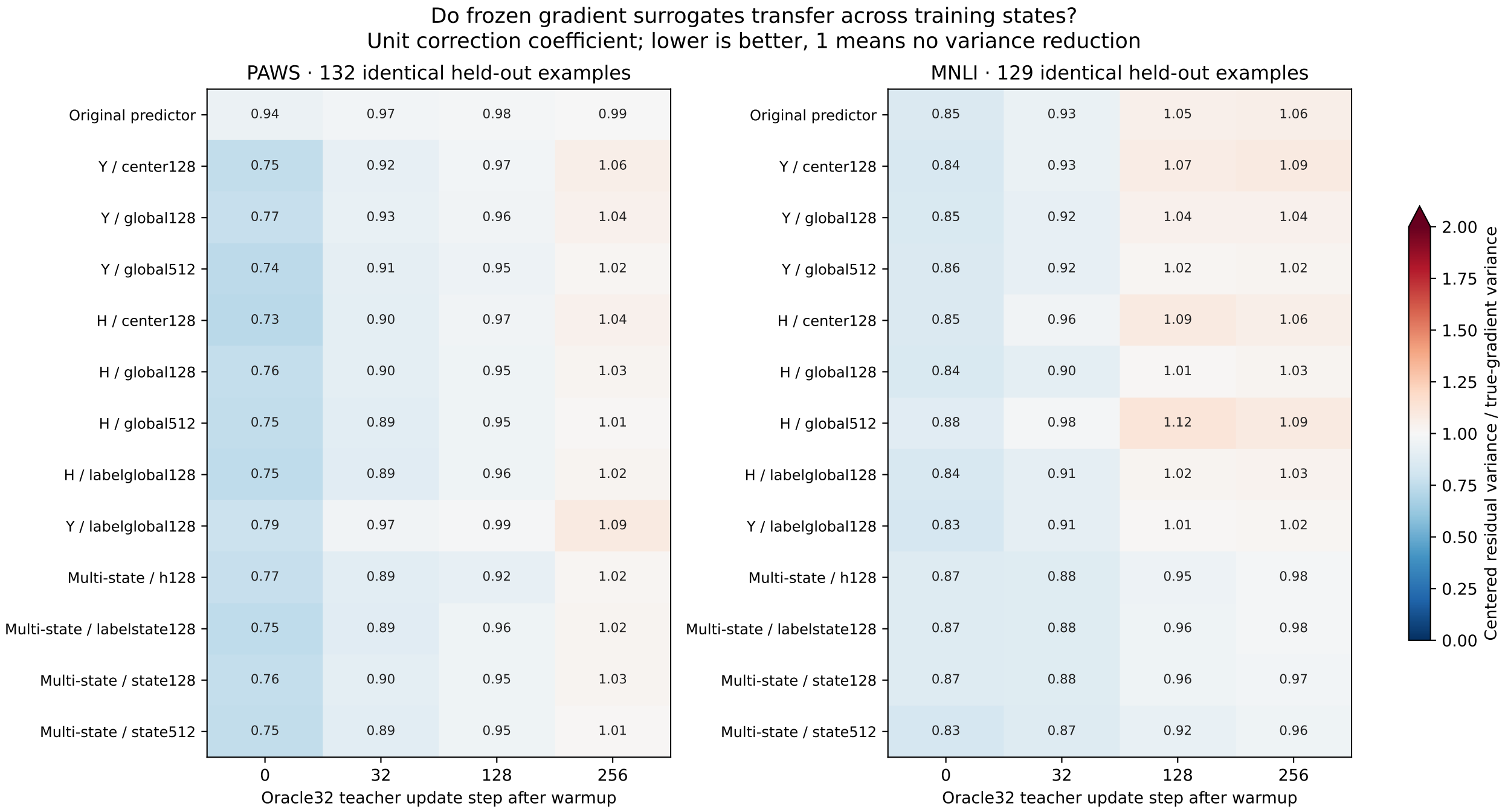
Do frozen gradient surrogates transfer across training states?
Unit correction coefficient; lower is better, 1 means no variance reduction
PAWS · 132 identical held-out examples
MNLI · 129 identical held-out examples
Original predictor
Y / center128
Y / global128
Y / global512
H / center128
H / global128
H / global512
H / labelglobal128
Y / labelglobal128
Multi-state / h128
Multi-state / labelstate128
Multi-state / state128
Multi-state / state512
0.94 0.97 0.98 0.99
0.75 0.92 0.97 1.06
0.77 0.93 0.96 1.04
0.74 0.91 0.95 1.02
0.73 0.90 0.97 1.04
0.76 0.90 0.95 1.03
0.75 0.89 0.95 1.01
0.75 0.89 0.96 1.02
0.79 0.97 0.99 1.09
0.77 0.89 0.92 1.02
0.75 0.89 0.96 1.02
0.76 0.90 0.95 1.03
0.75 0.89 0.95 1.01
0.85 0.93 1.05 1.06
0.84 0.93 1.07 1.09
0.85 0.92 1.04 1.04
0.86 0.92 1.02 1.02
0.85 0.96 1.09 1.06
0.84 0.90 1.01 1.03
0.88 0.98 1.12 1.09
0.84 0.91 1.02 1.03
0.83 0.91 1.01 1.02
0.87 0.88 0.95 0.98
0.87 0.88 0.96 0.98
0.87 0.88 0.96 0.97
0.83 0.87 0.92 0.96
0 32 128 256
Oracle32 teacher update step after warmup
Centered residual variance / true-gradient variance
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00